



Evolution of Pit Stop Strategies in Formula 1

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Why Pit Stops Matter





Research Problem & Motivation

- Pit stops are crucial in determining F1 race outcomes.
- Changes in regulations, car designs, and tire types from 2018–2024 have transformed pit strategies.
- The project seeks to understand and quantify these changes across seasons, teams, and tracks.

Research Question

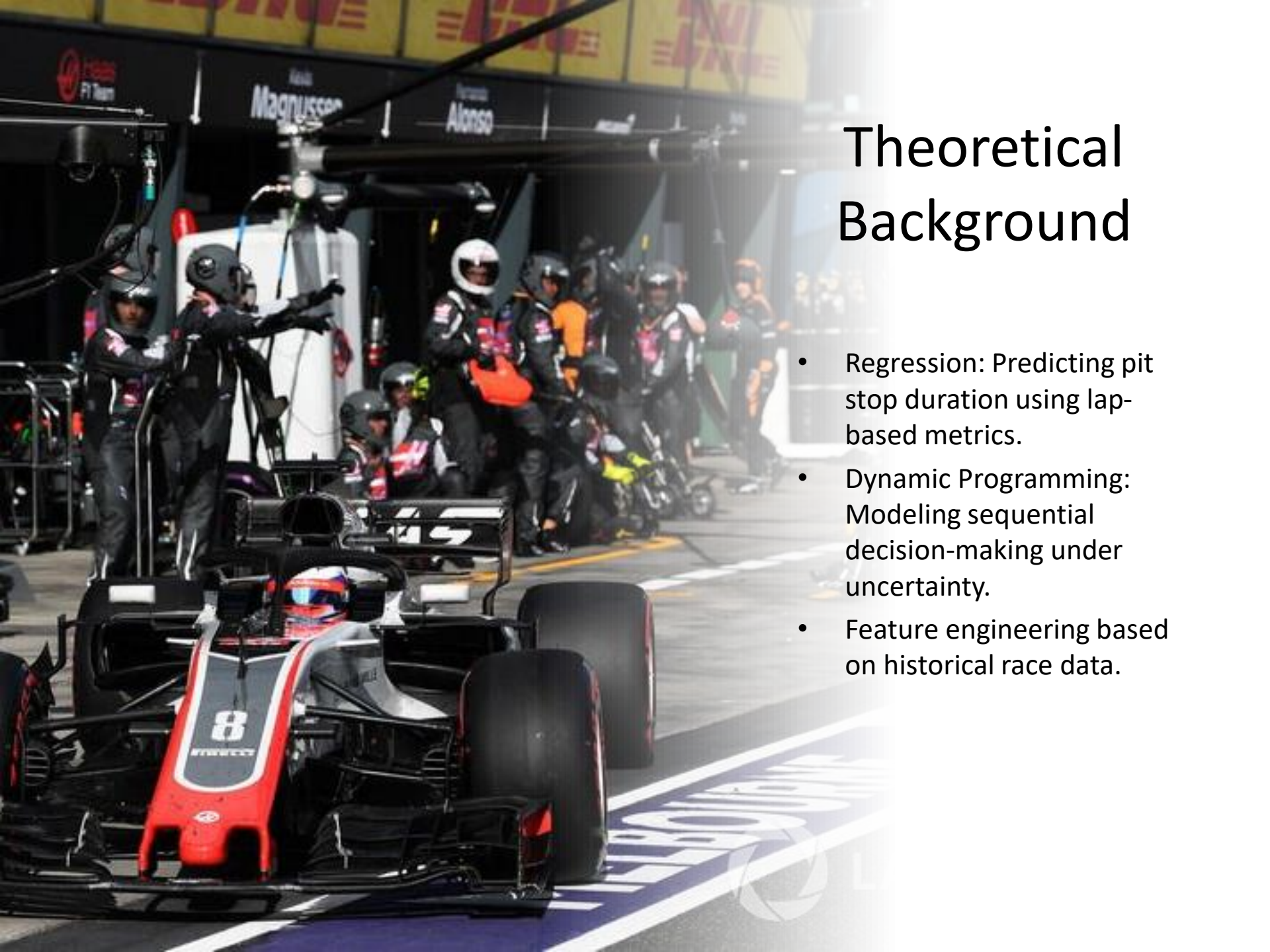
- How have Formula 1 pit stop strategies evolved between 2018 and 2024 in terms of:
 - Pit stop frequency
 - Pit stop duration
 - Tire compound usage...across different teams and circuits?



Literature Review

- **Optimization of Pit-Stop Strategies in Formula 1 Racing**
 - Utilizes regression-based modeling of tire wear and lap-time behavior.
 - Proposes a real-time decision framework for pit strategy.
(Carrasco & Thraves, 2022)
- **On the Optimization of Pit-Stop Strategies via Dynamic Programming**
 - Models uncertainty using stochastic dynamic programming.
 - Includes parameters like yellow flags, tire degradation, and track features.
(Carrasco & Thraves, 2023)
- These studies provide a theoretical foundation for our regression-based modeling and race strategy simulation.



A Formula 1 pit stop scene. In the foreground, a red and black Formula 1 car with the number 8 is visible. Several crew members in black and red uniforms are working around the car. In the background, other crew members and a white pit board are visible. The scene is set on a racetrack with a blue and white striped pit lane.

Theoretical Background

- Regression: Predicting pit stop duration using lap-based metrics.
- Dynamic Programming: Modeling sequential decision-making under uncertainty.
- Feature engineering based on historical race data.



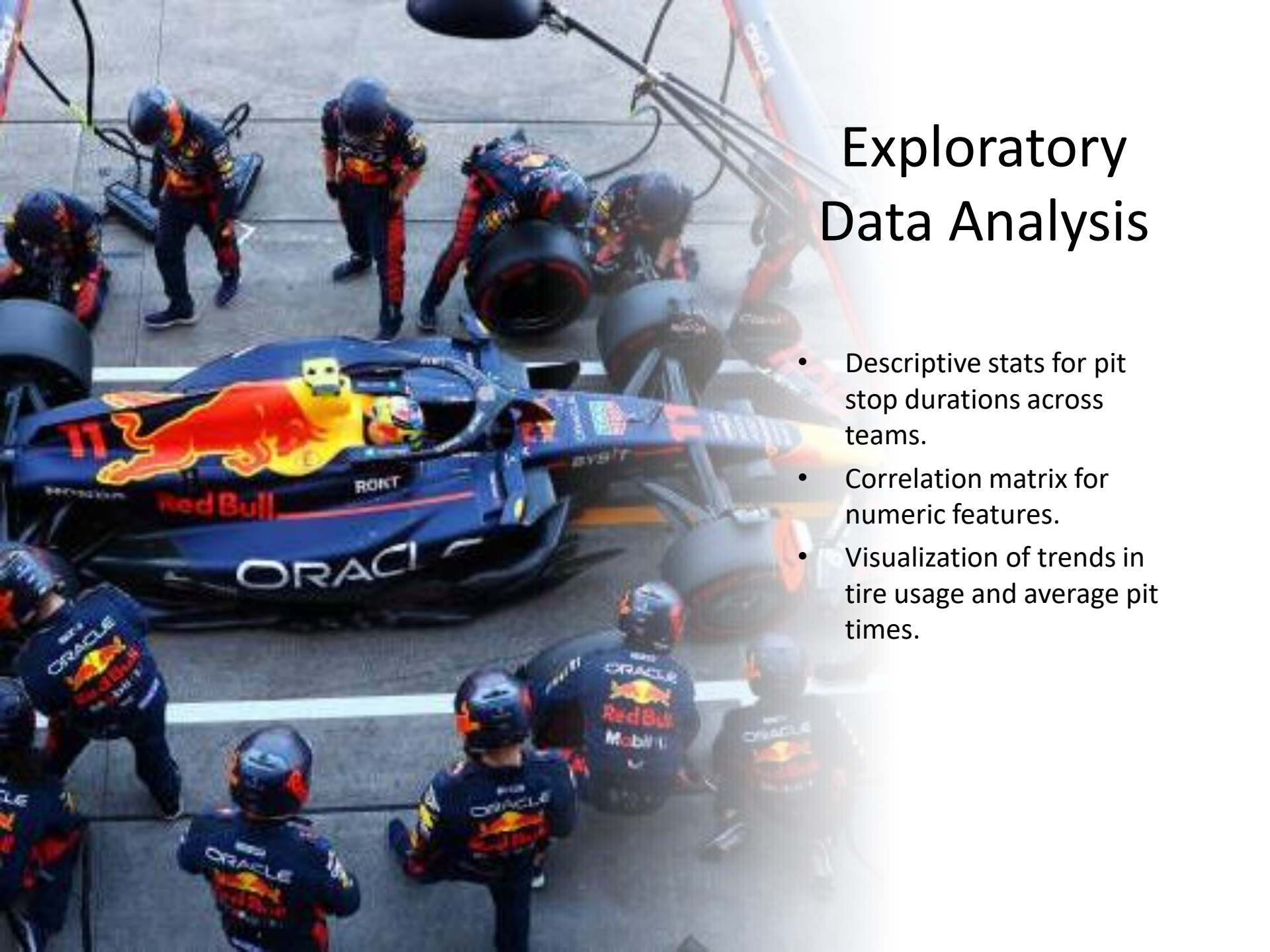
Research Design & Methodology

- Data: 2018–2024 F1 pit stop dataset (cleaned, filtered).
- Quantitative approach with regression modeling.
- Variables: Pit Stop Time (target), Tire Type, Team, Track, Lap Number, etc.
- Tools: Python (pandas, sklearn, seaborn, matplotlib)



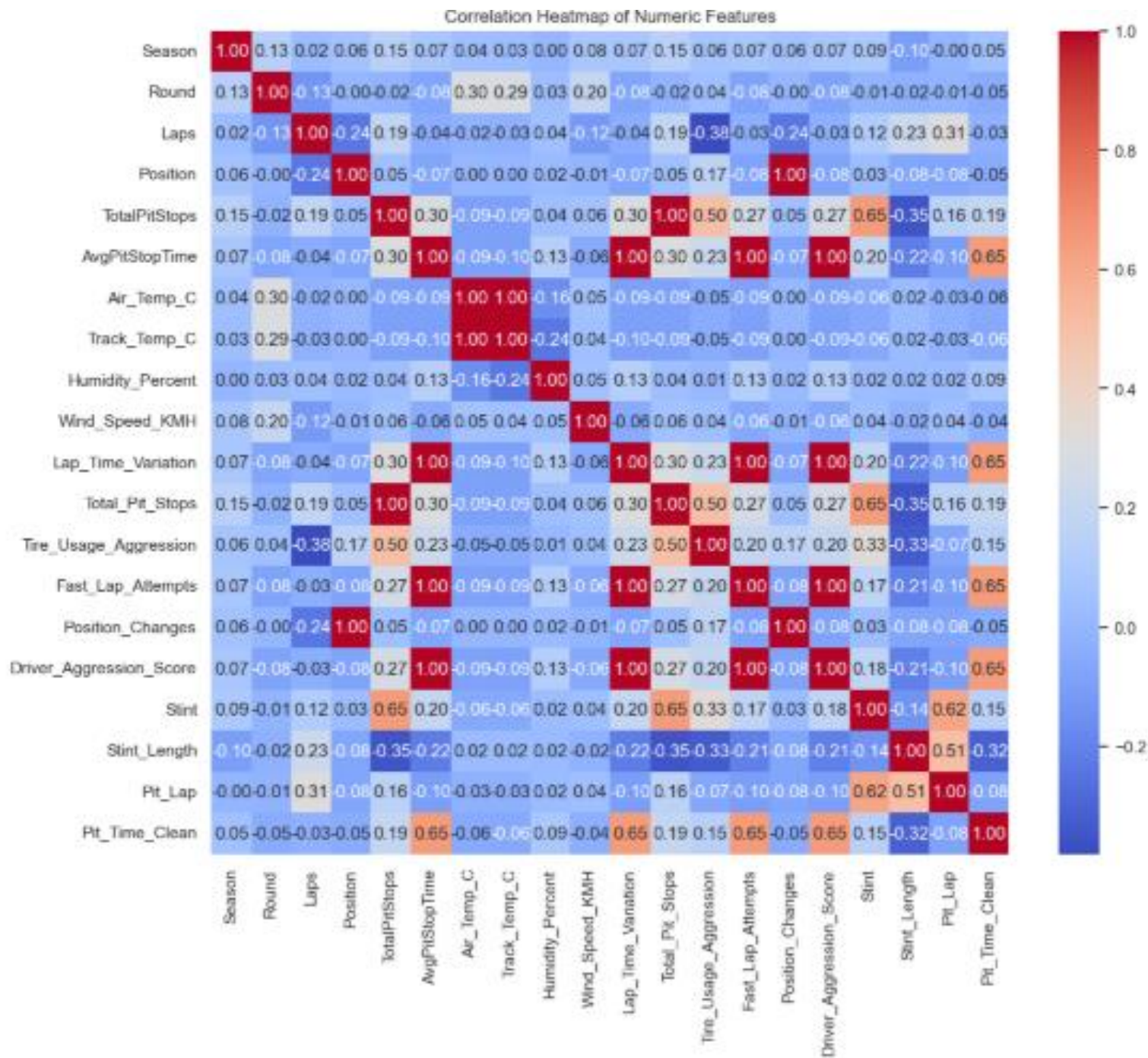
Data Preparation

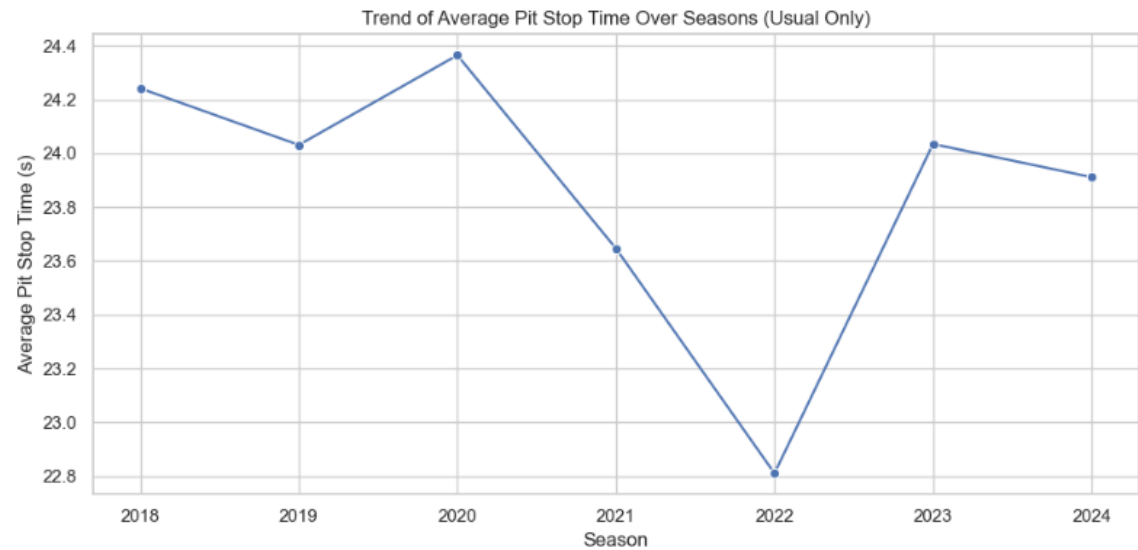
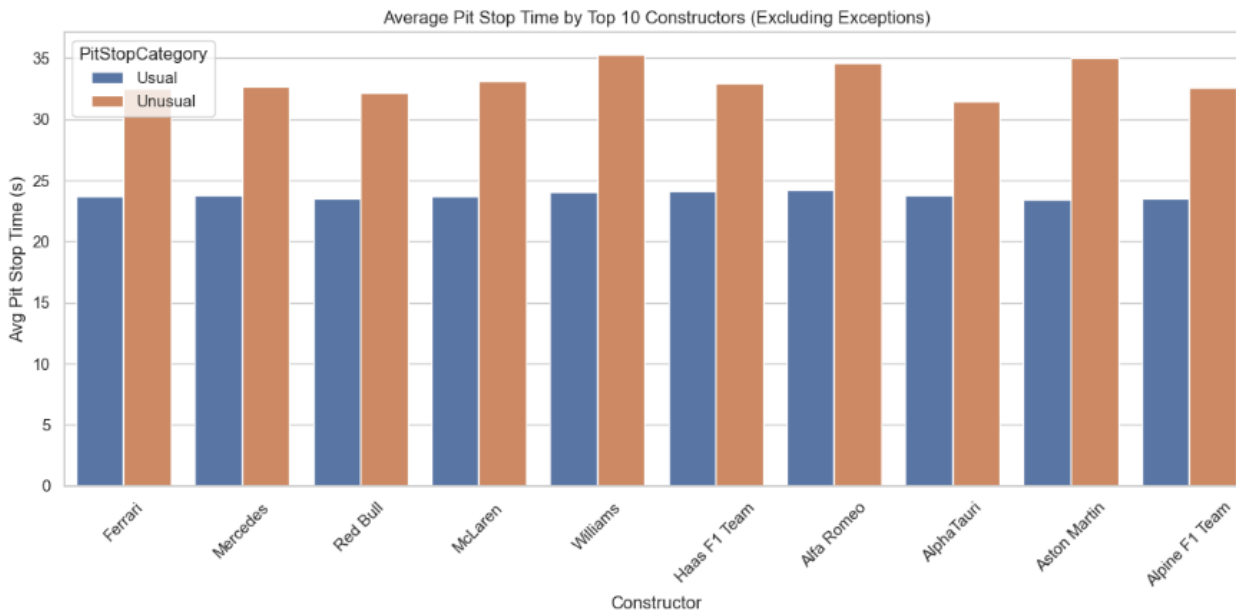
- Checked and handled missing values.
- Removed duplicates.
- Converted date columns, normalized numeric types.
- Data preprocessed for model input and visualization.

A high-angle photograph of a Red Bull Formula 1 car, number 11, being serviced by a pit crew. The car is dark blue with prominent red and yellow Red Bull branding. The word 'ORACLE' is visible on the sidepods. Several crew members in matching blue and red uniforms are positioned around the car, working on the tires. A crane arm is visible in the upper right corner. The scene is set on a paved track surface.

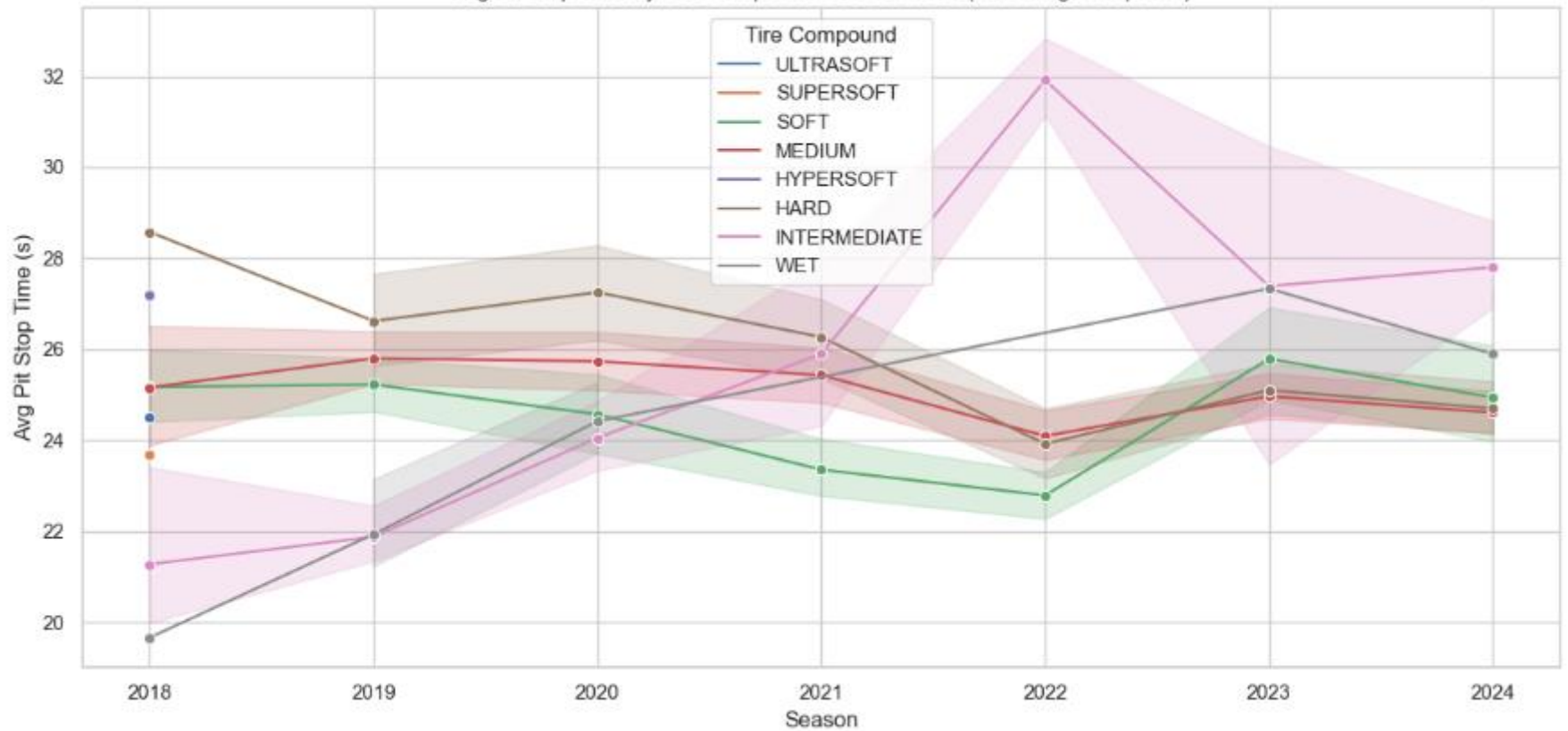
Exploratory Data Analysis

- Descriptive stats for pit stop durations across teams.
- Correlation matrix for numeric features.
- Visualization of trends in tire usage and average pit times.





Avg Pit Stop Time by Tire Compound Over Seasons (Excluding Exceptions)

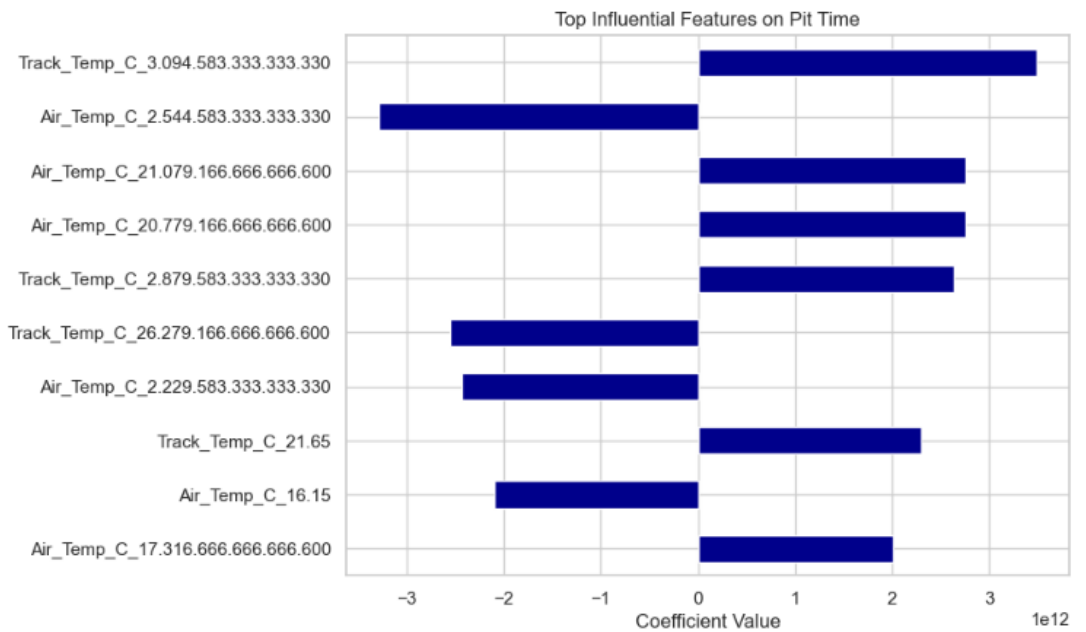


Regression Analysis

- Linear Regression applied to model pit stop time.
- Variables: Lap number, tire compound, team, track.
- Performance evaluated using R^2 score.
- Helps identify which factors most affect pit stop performance.



Feature importance





```
X = df_encoded.drop('Pit_Time', axis=1)
```

```
y = df_encoded['Pit_Time']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("R2 Score:", r2_score(y_test, y_pred))
```

```
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

```
R2 Score: 0.7402961111667633
```

```
RMSE: 1.4609561506556905
```

```
plt.figure(figsize=(8, 5))
```

```
sns.scatterplot(x=y_test, y=y_pred, alpha=0.6)
```

```
plt.xlabel("Actual Pit Time (s)")
```

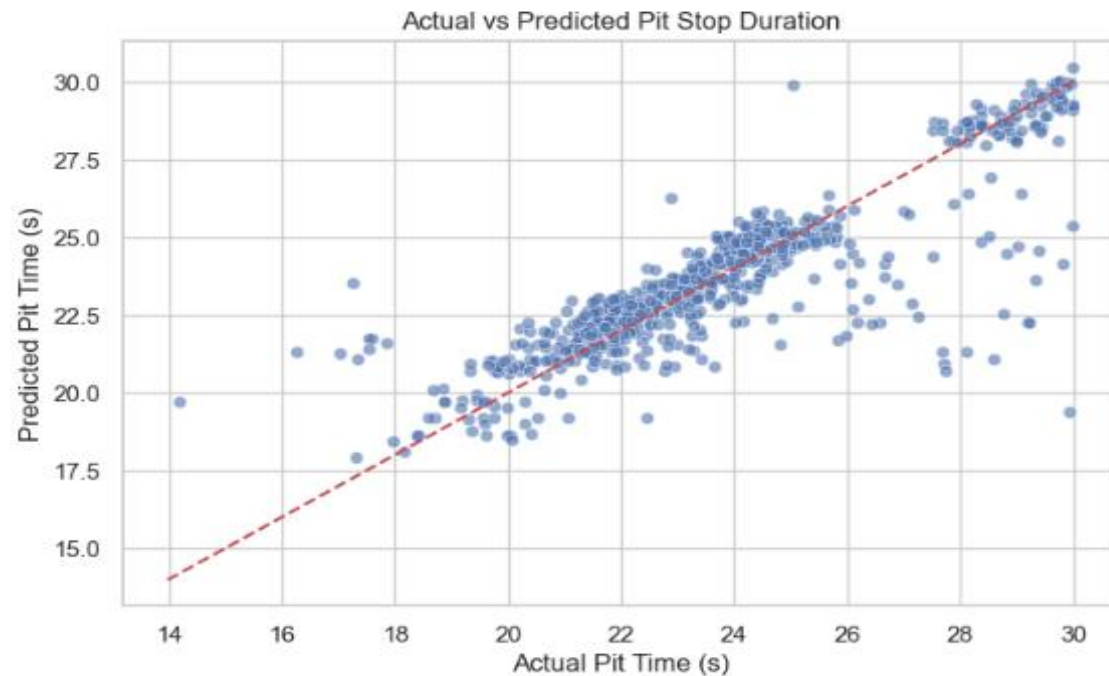
```
plt.ylabel("Predicted Pit Time (s)")
```

```
plt.title("Actual vs Predicted Pit Stop Duration")
```

```
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--') # ideal line
```

```
plt.grid(True)
```

```
plt.show()
```

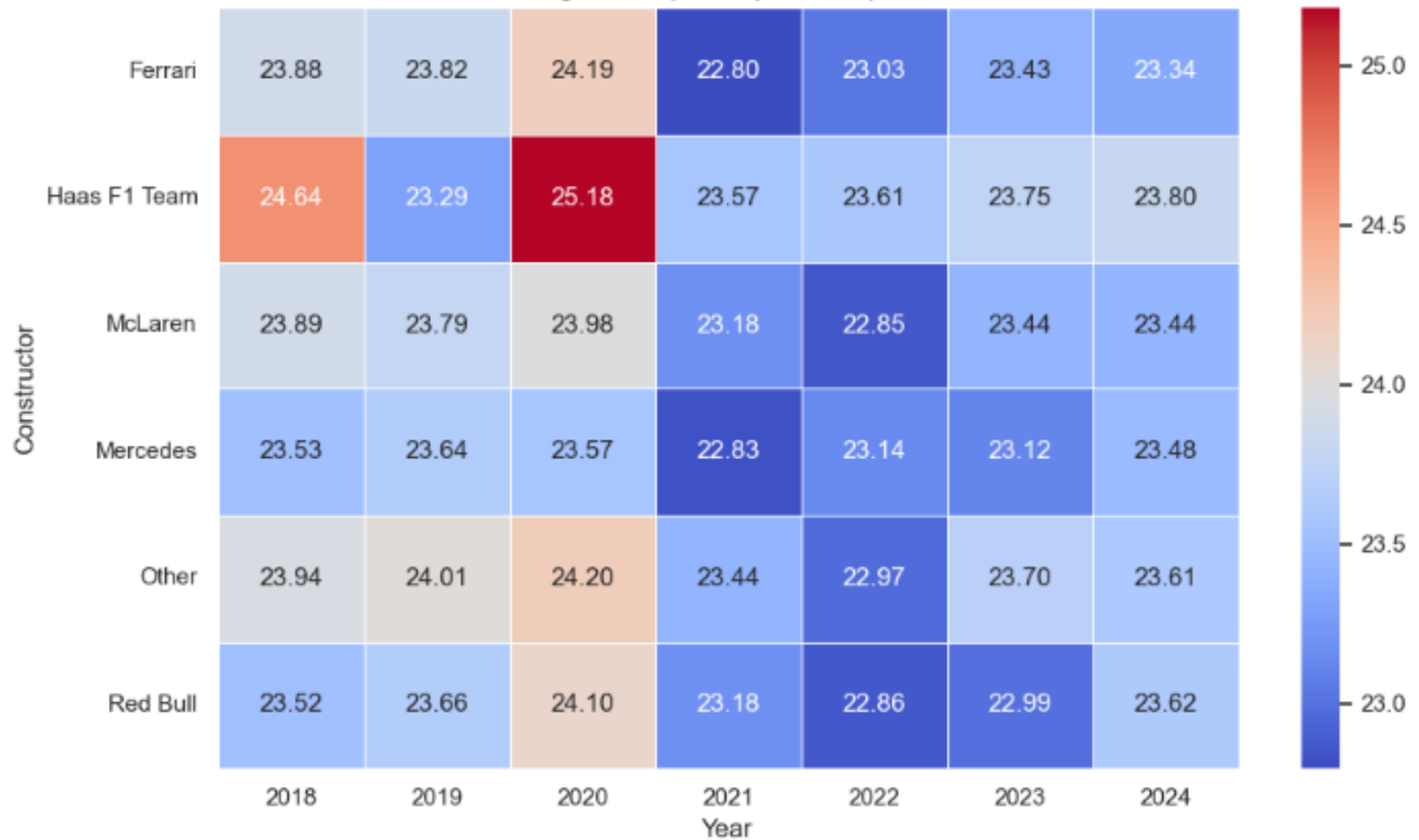




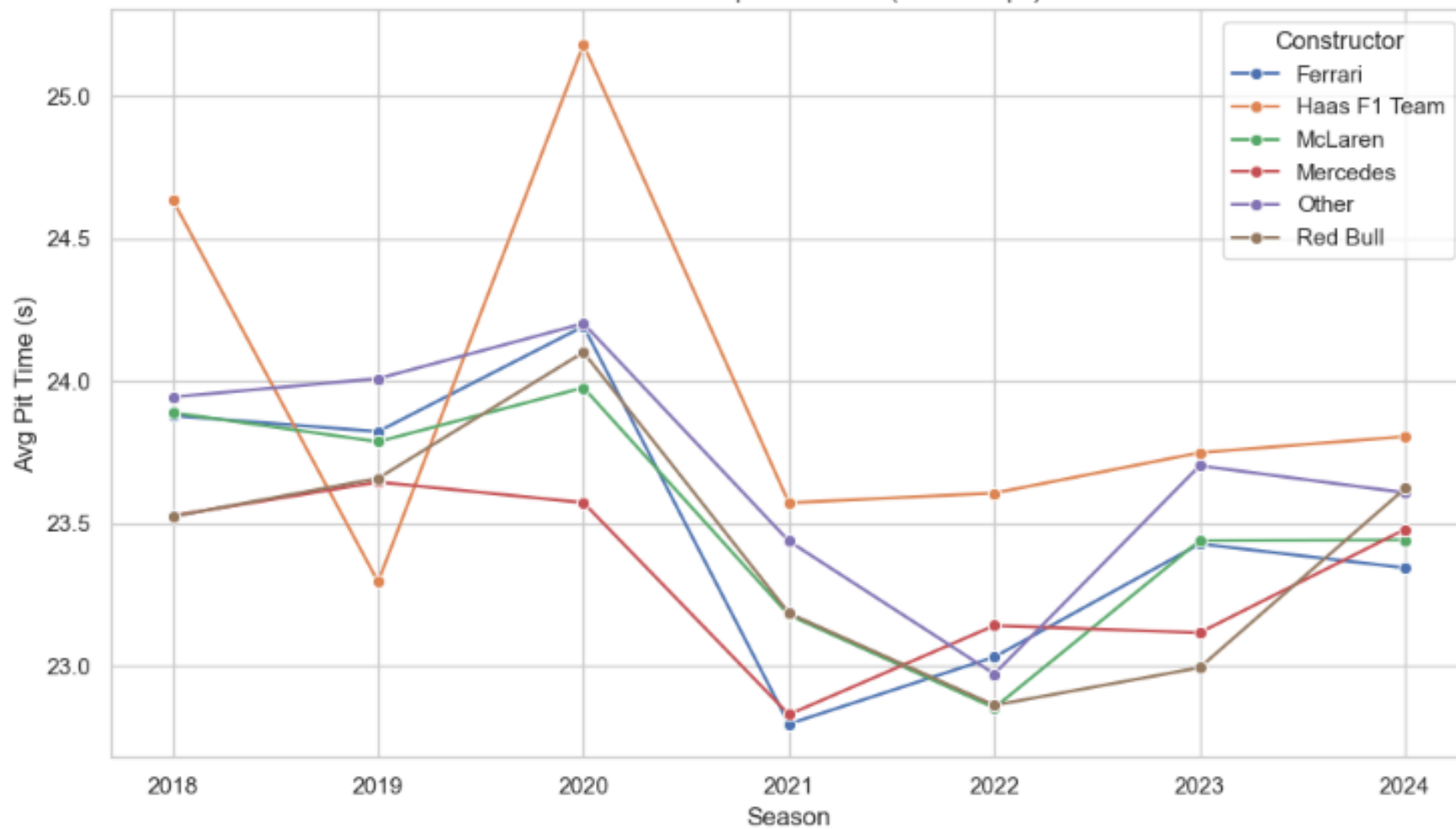
Results

- Notable trends in reduced pit time across top teams (e.g., Red Bull, Ferrari).
- Tire compound and lap number significantly affect pit durations.
- Strategies vary by track type and weather-related conditions.

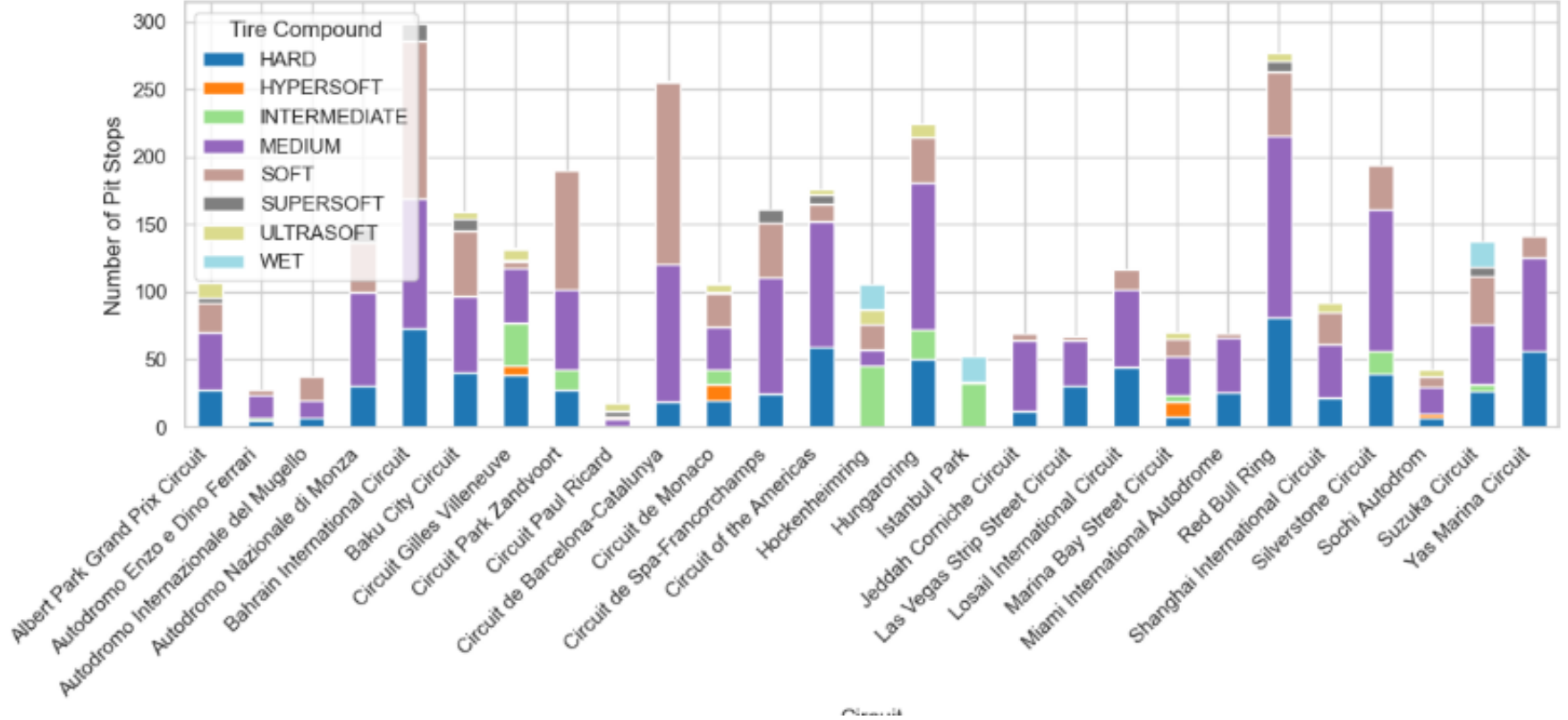
Average Pit Stop Time per Team per Year



Team-wise Pit Stop Time Trend (Usual Stops)



Tire Compound Strategy by Circuit



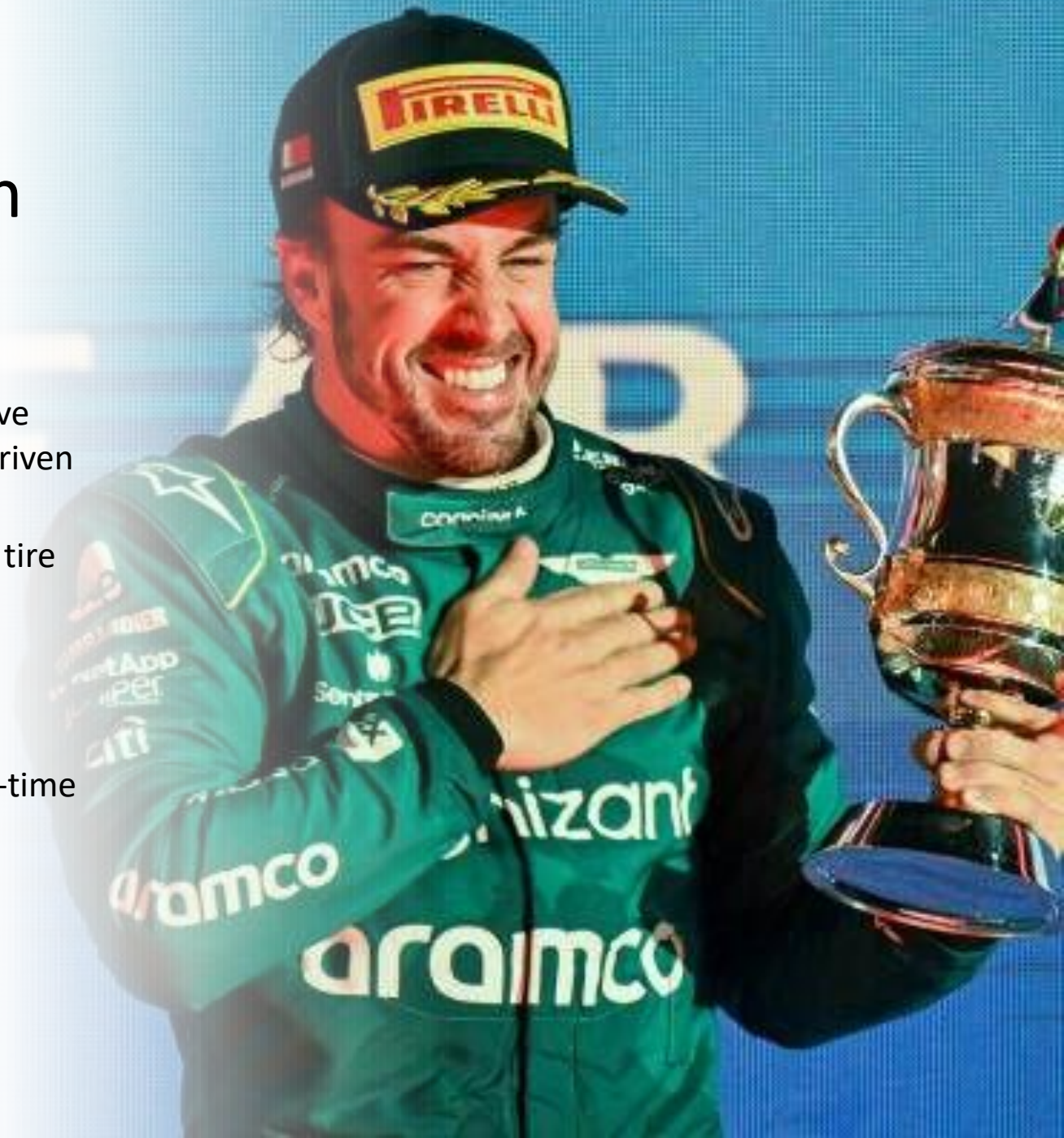
Interpretation & Discussion

- Findings align with dynamic decision theory from literature.
- Teams appear to adapt strategies based on regulation and performance constraints.
- Some anomalies suggest potential team-specific strategies worth deeper review.



Conclusion

- Pit stop strategies have become more data-driven over time.
- Key variables include tire compound and lap strategy.
- Teams continuously optimize based on competition and real-time variables.





Future Research

- Integrate weather and traffic data for enhanced prediction.
- Use machine learning models (XGBoost, LSTM) for better forecasting.
- Evaluate effect of rule changes across seasons.



References

- Carrasco, J., & Thraves, A. (2022). *Optimization of Pit-Stop Strategies in Formula 1 Racing*.
- Carrasco, J., & Thraves, A. (2023). *On the Optimization of Pit-Stop Strategies via Dynamic Programming*.
- F1 Pit Stop Dataset (2018–2024).
- Python Libraries: pandas, sklearn, seaborn, matplotlib.



Thank You!